

Plotting land parcels from the documents themselves

A technical report on the geometry layer: what has to be extracted from an Indian land record before anything can appear on a map, how the OCR stage hands over to the GIS stage, the algorithms that turn nineteenth-century field measurements into WGS84 polygons, and what is built today versus what an ideal system would do.

2,983

LINES OF ENGINE

0.025 m

BEST ASSEMBLY ERROR

20

INDIAN CRSS HANDLED

0

EXTERNAL MAP LAYERS

The one sentence that governs everything else

A 7/12 extract, a jamabandi row or a khatauni entry contains **no coordinates and cannot be plotted**. Geometry lives in a second, different document for the same survey number — the **tippa**n or **Field Measurement Book**. Every design decision in this report follows from that fact.

Prepared for	Team Brahmastra_1 · SIH 2026
Repository	D:\land\brain-land
Scope	Geometry & GIS layer. OCR is upstream and not yet built.
Status	Engine complete and tested; OCR contract defined and stubbed

The sample villages referenced throughout are worked examples of the input format. Their polygons are genuinely computed by the production engine from bearings, distances and offsets — nothing is a pre-drawn map — but the measurements were authored rather than lifted from a record room.

SECTION 1

Why a Record of Rights cannot be plotted

Almost every team approaching this problem statement assumes the pipeline is *scan the 7/12 → OCR it → put it on a map*. That pipeline cannot work, and it is worth being precise about why.

A Record of Rights is a **fiscal** document. Its job is to record who pays revenue on what, and it does that with a survey number, an owner, an area and a land classification. It has never carried coordinates, because the parcel's shape was recorded separately by the survey establishment, in the **tippan** (Maharashtra, Karnataka) or the **Field Measurement Book** (Tamil Nadu, Telangana, Andhra Pradesh). Two documents, two custodians, one survey number joining them.

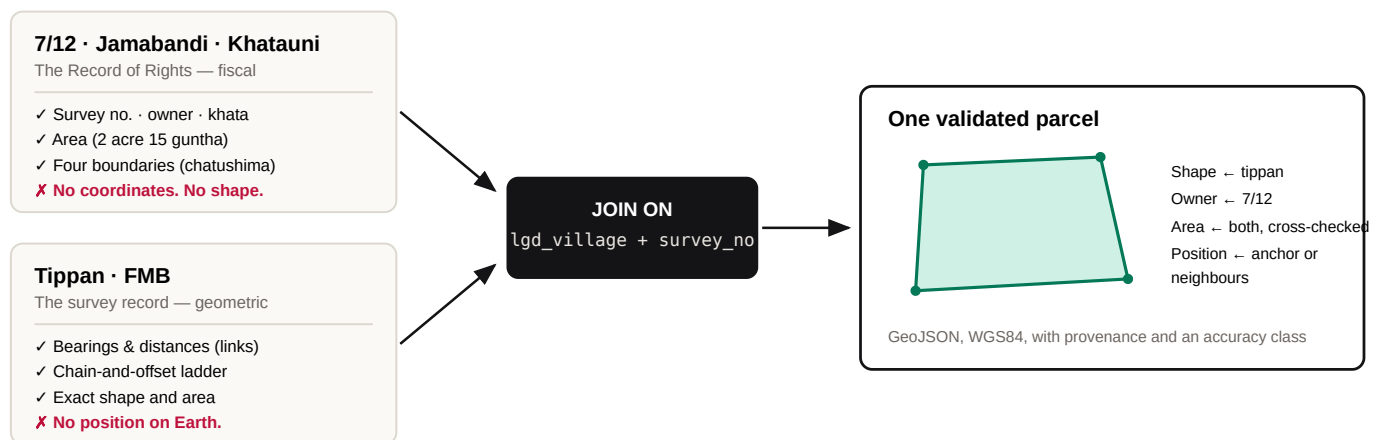


Figure 1. The two-document model. Neither record is plottable on its own; the survey number joins them. This is the single most important structural fact about Indian land records for a GIS project, and it determines what the extraction stage must target.

Practical consequence for the extraction plan

If the OCR effort targets only the 7/12 / jamabandi corpus, the system will produce a perfect searchable text database and **zero** map coverage. The tippan and FMB corpus — different shelves, different registers, often a different office — has to be in scope from the start, and it is harder: it is a hand-drawn diagram with a numeric table attached, not a ruled text table.

What each source can and cannot give you

TIER	SOURCE	SHAPE	AREA	POSITION	ORIENTATION	SELF-CHECK
A	Coordinates	EXACT	EXACT	EXACT	EXACT	NONE
B	Traverse	EXACT	EXACT	NONE	EXACT	CLOSURE
C	Chain & offset	EXACT	EXACT	NONE	NONE	NONE
D	Four boundaries	NONE	DECLARED	RELATIVE	NONE	NONE

Tier B is the only source that validates itself: a closed boundary walk must return to its start, and the distance by which it fails is a direct measure of transcription quality. Tier D is the only source present on essentially every document, and although it gives nothing alone, across a village it becomes the constraint network that positions everything (Section 6).

SECTION 2

What must be extracted, exactly

This is the contract the OCR stage has to satisfy. Anything short of it produces a record that can be searched but not mapped.

2.1 Mandatory on every document, whatever its type

FIELD	TYPE	WHY THE GEOMETRY LAYER NEEDS IT
survey_no	string	The join key: RoR ↔ tippan, and parcel ↔ neighbour. Must preserve the sub-division (hissa) exactly — <code>124/2/A</code> is not <code>124/2</code> .
village, tehsil, district, state	string	Identity, and the state decides regional unit values. Resolve to LGD codes where possible so spelling variants collapse.
area_value	number	The independent check on every reconstruction. Without it, a geometry can be wrong and nothing will notice.
area_unit	string	guntha, acre, hectare, kanal, bigha... Regional units additionally require the state.
boundaries	object	north / south / east / west. Feeds the adjacency solver. Capture the raw text — "Survey No. 125" and "Nala" are both meaningful.

2.2 Then whichever geometry source the document carries

Tier A — Coordinates

Found in: resurvey / DGPS records, SVAMITVA property cards, modern FMB sheets, deed schedules that quote coordinates.

coordinates.points[] Boundary vertices in order, as `[x, y]` pairs.

coordinates.crs **Not optional.** "WGS84", a UTM zone, or a full India zone name. See Section 5.2.

coordinates.order `xy` or `yx` — whether the sheet writes long,lat or lat,long.

Tier B — Traverse

Found in: Field Measurement Book, tippan, survey field books, deed schedules given as bearings and distances.

traverse.legs[].bearing Whole circle (`125-30-00`) or quadrantal (`N45°30'E`). Both notations must round-trip.

traverse.legs[].distance Numeric.

traverse.legs[].unit links and chains are the usual units in old books; also m, ft, karam, gaz.

magnetic_declination_deg Only if the bearings are magnetic. Omitting it when they are rotates the whole parcel.

Tier C — Chain and offset (the tippan "ladder")

Found in: tippan sketches, village measurement books. This is the commonest geometric record in the legacy corpus and the hardest to extract, because it is a drawing with numbers on it rather than a table.

chain_offset.base_length Length of the base line the surveyor ran across the field.

offsets[].chainage Distance along the base line. **May be negative** — a corner can lie behind the starting station.

offsets[].offset Perpendicular distance to the boundary point.

offsets[].side **L** or **R** of the base line. Getting this wrong mirrors the parcel.

offsets[].seq Corner order, if the sketch numbers its points. Removes all ambiguity in ring construction.

base_bearing True bearing of the base line, if stated. Without it the shape is exact but its orientation is unknown.

Tier D — Four boundaries

`boundaries.north / south / east / west`, captured as raw text. The solver distinguishes a parcel reference ("Survey No. 125", "S.No 124/2/A") from a non-parcel edge ("Nala", "Village road", "Government land", "Kaluva") and uses the former as constraints. Non-parcel edges are retained, not discarded — they tell you the parcel is on the edge of the block.

2.3 Optional, but each one materially improves the result

control_points[] Any corner whose real position is known: a trijunction pillar, a GTS benchmark, or simply two corners an officer clicks on satellite imagery. **Two of them fix position, rotation and scale simultaneously.**

parent_survey_no Enables the subdivision-sum check — children must tile the parent.

village_map_sheet_no Which cadastral sheet the tippan came from; groups parcels for batch assembly.

survey_date Magnetic declination drifts over decades. An 1890 compass bearing and a 1970 one need different corrections.

The machine-readable version of this whole section is served live at `GET /api/gis/requirements`, so the extraction team can code against it rather than against a document.

SECTION 3

The OCR → GIS handoff, in detail

The geometry engine and the OCR stage are deliberately decoupled by a single JSON contract. The engine has never seen an image and never will; OCR never has to know what a Lambert Conformal Conic projection is. This is why the GIS layer is finished while OCR has not started.

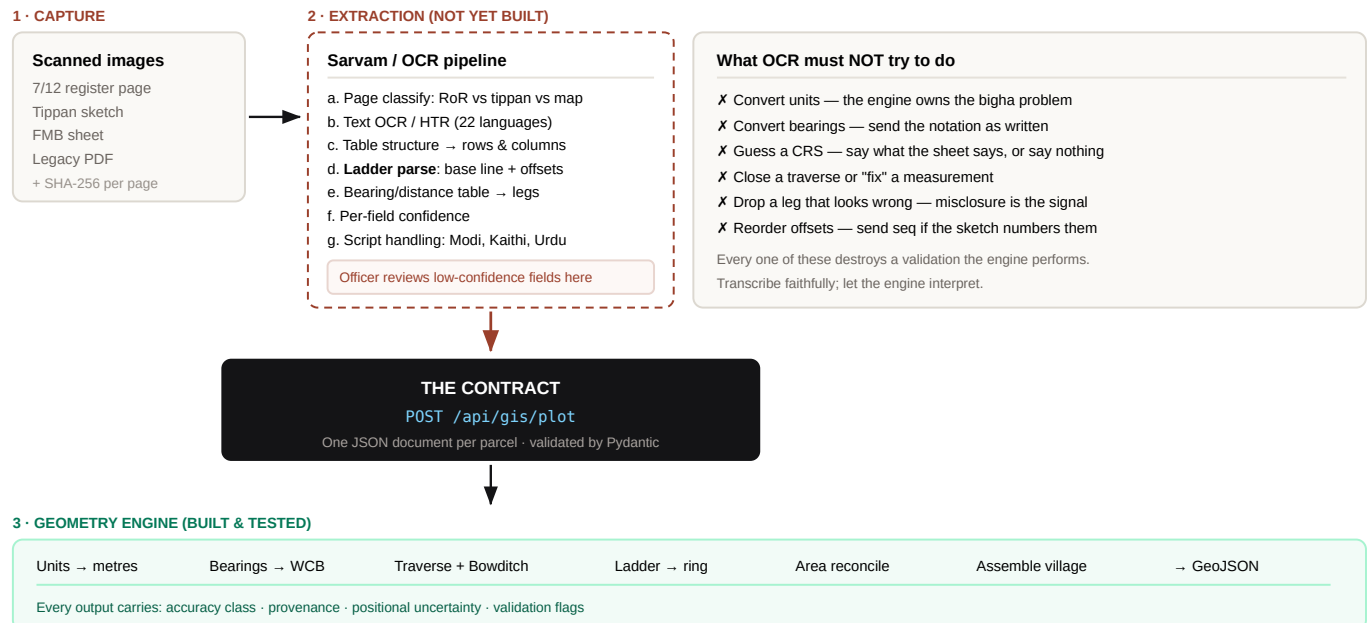


Figure 2. The handoff. The dashed box is the only part not yet built. Because the boundary is a validated JSON schema rather than a function call, the geometry side was developed and tested to completion against hand-authored documents.

3.1 The contract, concretely

```
// One object per parcel. POST an array of them to /api/gis/plot
{
  "survey_no": "201/5",           // join key — preserve hissa exactly
  "village": "Hosahalli", "tehsil": "Mysuru",
  "district": "Mysuru", "state": "Karnataka",
  "lgd_village_code": "613204",
  "owner_name": "Siddaraju H M",
  "area_value": 124.29, "area_unit": "guntha", // the independent check

  "boundaries": {                 // Tier D — raw text, always send it
    "east": "Survey No. 201/6", "west": "Survey No. 201/4",
    "south": "Survey No. 201/2", "north": "Survey No. 201/8"
  },

  // Then ONE of the following three, whichever the document has:

  "coordinates": { "points": [[76.639,12.295], /*...*/],
    "crs": "WGS84", "order": "xy" },

  "traverse": { "legs": [
    { "bearing": "N85°45'E", "distance": 571.8, "unit": "links" },
    { "bearing": "S02°30'E", "distance": 522.1, "unit": "links" } /*...*/
  ], "distance_unit": "links", "adjustment": "bowditch" },

  "chain_offset": { "base_length": 572.2, "unit": "links",
    "offsets": [
      { "chainage": 0.0, "offset": 0.0, "side": "L", "seq": 0 },
      { "chainage": 572.2, "offset": 0.0, "side": "L", "seq": 1 },
      { "chainage": 619.7, "offset": 522.1, "side": "L", "seq": 2 },
```

```

    { "chainage": -28.1, "offset": 497.3, "side": "L", "seq": 3 } // negative is legal
  ] },

  "control_points": [
    { "local": [0,0], "lon": 76.6394, "lat": 12.2958, "label": "SW pillar" }
  ]
}

```

3.2 The eight rules the extraction stage must follow

#	RULE	REASON
1	Transcribe, do not interpret	Send "N45°30'E", not 45.5. The engine parses both notations and records which was on the page.
2	Never convert units	Send the number and the unit written. Unit conversion is state-dependent and the engine refuses unsafe cases rather than guessing.
3	Never repair a measurement	A leg that looks wrong <i>is</i> the finding. Closure analysis detects it; silently fixing it destroys the detector.
4	Preserve order and sequence	Legs in traverse order; offsets with seq when the sketch numbers corners. Reordering changes the polygon.
5	State the CRS or omit it	A guessed CRS is worse than none. Section 5.2 quantifies this at 195 km.
6	Send all four boundaries verbatim	Including "Nala" and "Village road". Non-parcel edges are information, not noise.
7	Emit per-field confidence	Routes low-confidence fields to officer review before geometry is attempted.
8	Batch by village, not by page	The adjacency solver needs the whole village at once; parcels georeference each other.

3.3 What happens to a document that fails

Nothing is silently dropped. A document that carries no usable source still returns a record, with `method: "none"`, an accuracy class, and a flag naming exactly what is missing and what to do about it — for example *"The document carries no coordinates, no traverse, no chain-and-offset measurements and no boundary description. Locate the tippan / Field Measurement Book for this survey number — the textual extract alone never contains geometry."* That flag is the work queue for the record room.

SECTION 4

System architecture

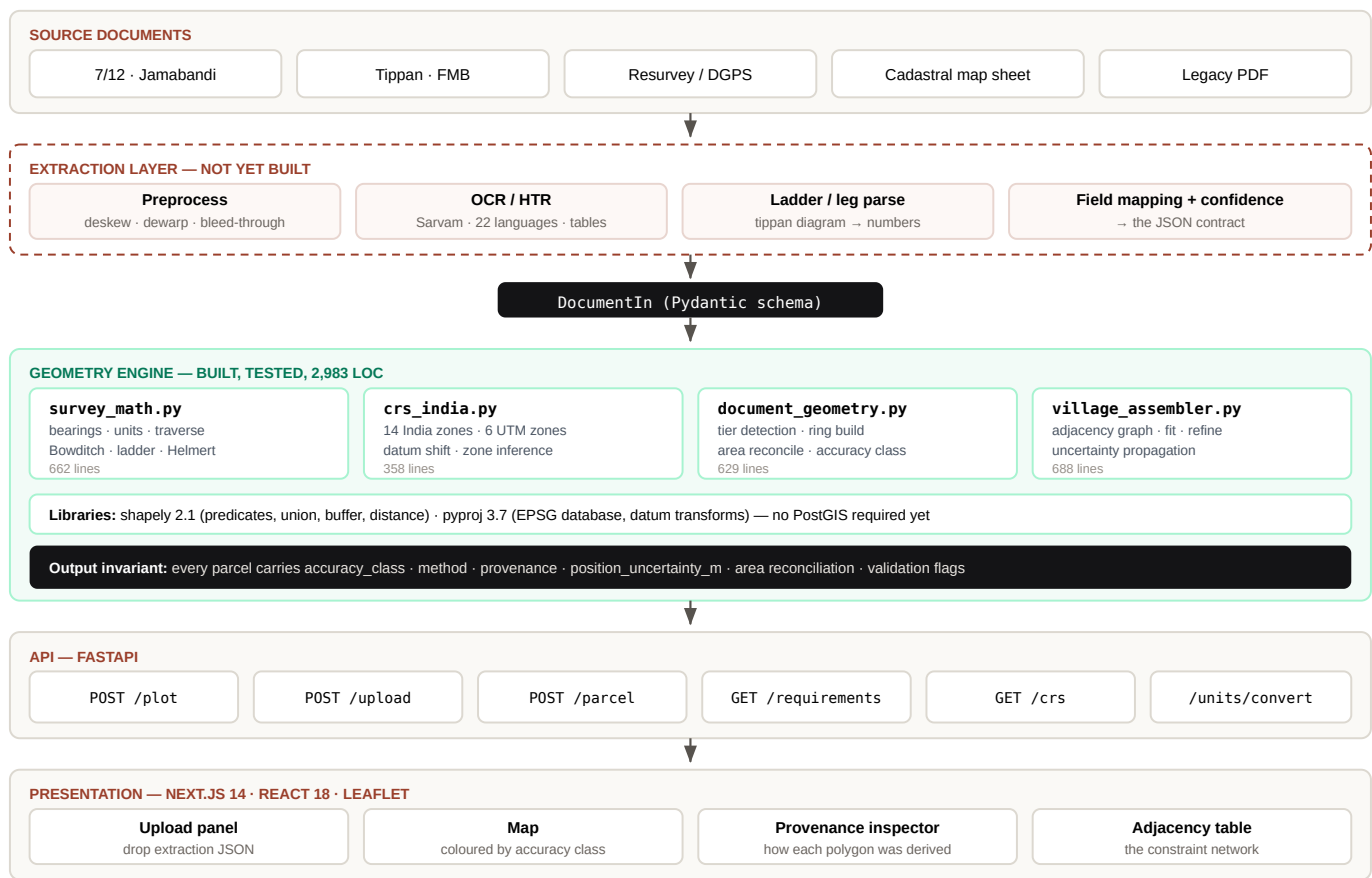


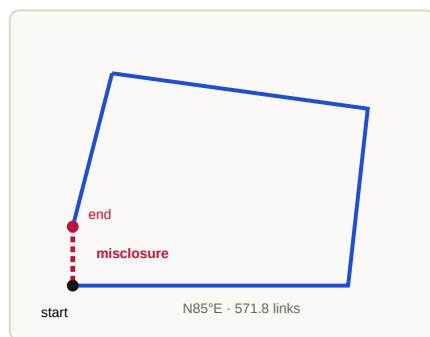
Figure 3. Layered architecture. The dashed layer is the only unbuilt one. The Pydantic schema between layers 2 and 3 is the contract that let the geometry work finish independently.

How the engine works

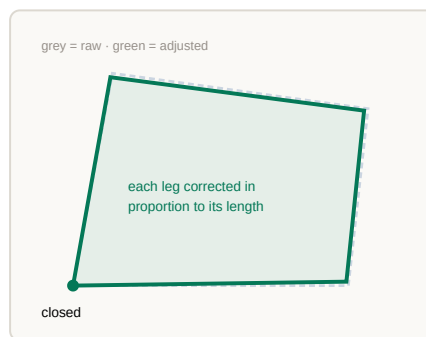
5.1 Traverse reconstruction and the closure check

A traverse is a walk round the boundary: from each corner, a bearing and a distance to the next. Each leg becomes a displacement — $\Delta E = L \cdot \sin(\theta)$, $\Delta N = L \cdot \cos(\theta)$ — and the walk should return to where it began. It never quite does. The distance by which it fails is the **misclosure**, and it is the single most valuable number in the whole document, because it detects transcription errors before anything is plotted.

a. The walk does not close



b. Bowditch distributes the error



c. The grade is the error detector

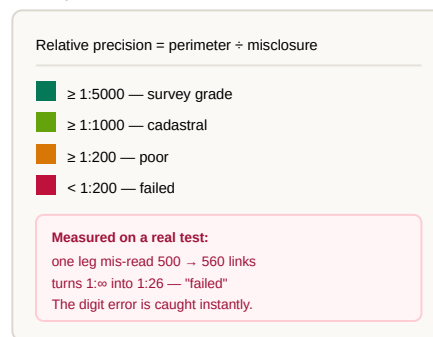


Figure 4. Traverse, adjustment and grading. The Bowditch (compass) rule corrects each leg in proportion to its length — the right choice when angles and distances are of similar reliability, as in a chain-and-compass cadastral survey.

5.2 Coordinate reference systems — why this is not a detail

Legacy Indian cadastral sheets are not on WGS84. They are on Everest 1830 / Kalianpur datums, in the Survey of India's **India zones** — Lambert Conformal Conic bands with standard parallels at 39°30', 32°30', 26°, 19° and 12° N.

The cost of getting it wrong, measured

A parcel at Pune (18.5089 N, 73.9259 E) on Kalianpur 1975 / India zone IIIa has easting 2,102,832 m, northing 871,056 m.

Read correctly	18.5089 N, 73.9259 E — Pune
Read as UTM 43N / WGS84	7.6345 N, 89.3833 E — the Bay of Bengal
Read on the wrong Kalianpur datum	195 km away

The engine builds its CRS registry from pyproj's EPSG database at import rather than hard-coding it, so it cannot drift from the authority. It covers all 14 India zones EPSG publishes, all six UTM zones spanning the country (42N–47N), and EPSG:7755 (WGS 84 / India NSF LCC) for national work. It selects the correct UTM zone per parcel from that parcel's own longitude, and can infer which legacy zone an undated sheet most likely used from the EPSG bounding boxes. Round-trip fidelity measured across Pune, Delhi, Kolkata and Madurai is 0.2–1.1 mm. It **refuses** an ambiguous name: "India zone IIIa" exists on both Kalianpur 1880 and 1975 and they are hundreds of metres apart.

5.3 The chain-and-offset ladder

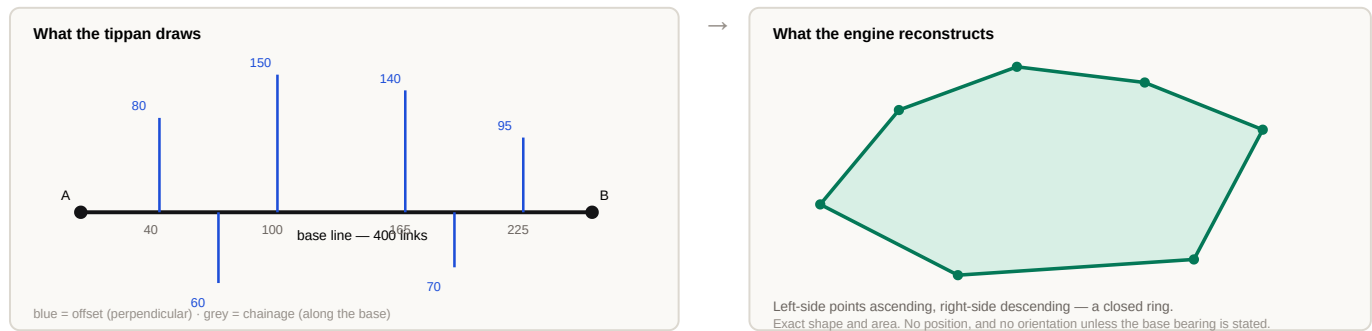


Figure 5. The "ladder to map" method. The base line becomes the local x-axis and each offset a y-coordinate. Points are ordered up the left side and back down the right so the ring traces the boundary once without self-intersecting; if the sketch numbers its corners, that order is used instead.

5.4 The accuracy class — the output invariant

Every parcel leaves the engine labelled with how its geometry was obtained. A parcel positioned by solving the adjacency network must never look, in the database or on the map, like one measured with a total station.

CLASS	MEANING	POSITION ERROR
SURVEYED	Coordinates taken directly from the document	Source accuracy
RECONSTRUCTED_ANCHORED	Shape rebuilt from measurements, placed using control points	Control point accuracy
RECONSTRUCTED_FLOATING	Shape and area exact; not yet placed on the ground	Not placed
INFERRED	Position estimated from adjoining parcels and declared area	Metres — reported per parcel
NONE	Document carries nothing geometry can be derived from	—

How a village assembles itself

This is the part that makes "everything from the documents, nothing external" actually work. A tippan gives an exact shape but no position. A 7/12 gives an exact area and the four boundaries but no shape. Neither is plottable alone — but across a village they form a constraint network, and **one anchored parcel places all of them**.

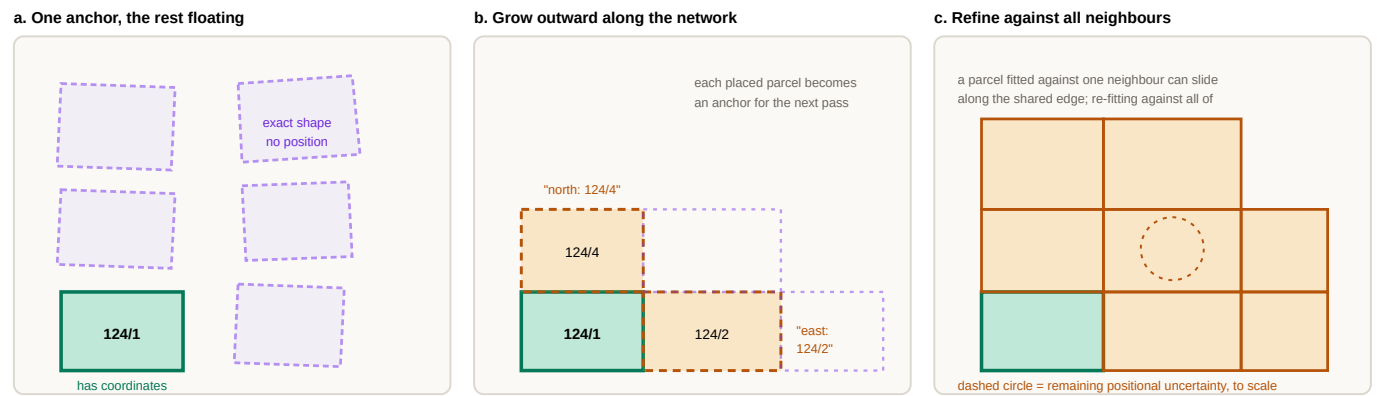


Figure 6. Village assembly. Growth passes place every parcel that has at least one mapped neighbour; refinement passes then re-fit each one against all of its neighbours simultaneously, which is what removes the slide-along-the-edge ambiguity.

6.1 The cost function

Placement is found by minimising a cost, expressed entirely in square metres so the terms are commensurable:

TERM	RATIONALE
Overlap × 12	Two parcels cannot occupy the same ground. Weighted heavily because a placement asserting they do is simply wrong.
Gap²	The record says these parcels touch; a gap is a violation.
(1 – cos θ) × L² × 0.5	Directional term. If the record says the neighbour is west, the parcel must sit east of it.
– shared edge fraction × L² × 0.30	Reward. Adjoining parcels do not merely touch at a point; they share a whole edge because one was surveyed off the other. Bounded — an unbounded version lets a parcel slide along its neighbour collecting contact, which made results worse.

Optimisation is a deterministic compass (pattern) search on translation, wrapped in a rotation sweep only when orientation is genuinely unknown — a tippan with no stated base bearing. A traverse of true bearings already knows where north is, and rotating it would discard real information.

6.2 Positional uncertainty

Each placement is probed in eight directions to find how far the parcel can move before the fit measurably worsens — a parcel constrained on three sides cannot move; one with a single neighbour can slide. That local slack is then **propagated along the inference chain**, because a parcel placed off a parcel placed off the anchor inherits its parent's error, exactly as in any survey network. A calibration factor of 2.5 is applied to the local term, validated against ground truth so the figure over-reports rather than under-reports. An error bar that is confidently too small is worse than none.

SECTION 7

Validation and measured results

7.1 The ground-truth test

A village is constructed with known geometry, converted into documents containing only bearings, distances, areas and boundaries, every position but one is discarded, and the engine is asked to rebuild it. The reconstruction is then compared against the original. Run it with `python tests/test_assembly_truth.py`.

SCENARIO	PARCELS	ANCHORS	PLACED	WORST CENTROID ERROR	AREA ERROR
Regular grid, all traverse	6	1	6/6	0.025 m	0.000 m ²
Irregular quadrilaterals, 2 rotation-unknown tippans	8	1	8/8	≈16 m	<0.03 m ²

Why the two numbers differ, and why that is honest rather than a defect

When parcels tile regularly the constraint network is over-determined — each parcel is hemmed in on several sides and only one placement satisfies everything, so the fit is centimetre-exact. Irregular quadrilaterals leave genuine slack: several placements satisfy "touches on the correct side" almost equally well. **Shape and area remain exact in both cases; only position degrades.** Rather than tune the optimiser until a synthetic case looks good, the system measures the slack and reports it per parcel.

7.2 Component tests performed during development

COMPONENT	TEST	RESULT
Bearing parser	20,016 random bearings, quadrantal ↔ whole-circle round trip	0 failures, worst error 0.05"
Unit conversions	Known identities: 40 guntha = 1 acre, 8 kanal = 1 acre, 100 links = 1 chain = 66 ft	All exact
Traverse	Perfect 100×50 m rectangle	5000.0000 m ² , exact closure
Error detection	One leg mis-transcribed 500 → 560 links	1:26, graded "failed"
Chain & offset	Both-sided and one-sided ladders	3000.00 / 2000.00 m ² exact
CRS round trip	WGS84 → legacy India zone → WGS84, four cities	0.2–1.1 mm
Helmert placement	Rotate 90°, scale 2 — area should quadruple	10,000 → 40,000 m ²
Scale sanity check	Control points 2.67× too far apart	Flagged, area tracks the fit
API	All endpoints, valid and malformed input	Correct codes and messages
Frontend	<code>tsc --noEmit</code> under strict mode	Clean

7.3 Three real bugs found by testing, and what they teach

BUG	CAUSE	LESSON
Tippan documents silently produced no geometry	<code>o.get("unit", default)</code> returns <code>None</code> when the key exists with a null value, so unit conversion failed	Serialised optional fields arrive as explicit nulls. Use falsiness, not <code>.get</code> defaults.
Negative chainage rejected as invalid	A validator asserted <code>chainage ≥ 0</code>	A boundary corner can legitimately sit behind the base line's start station. Real tippans record this.
Whole villages failed to assemble	The neighbour regex required a leading digit, so non-numeric survey numbers never entered the adjacency graph	Match against the known survey numbers first. A parcel missing from the network is a parcel that never gets placed.

A fourth was caught by a formatting test: rounding 0.0999° to degrees-minutes-seconds produced `0°05'60.0"` instead of `0°06'00.0"` — the classic carry bug, fixed by rounding to tenths of a second before splitting.

SECTION 8

Technology stack

LAYER	TECHNOLOGY	WHY THIS CHOICE
Extraction TO BUILD	Sarvam Document Digitisation	Indian sovereign AI, 22 scheduled languages, handles merged cells and multi-level headers — which is what a ruled revenue register is
	OpenCV, DocTR	Deskew, dewarp, bleed-through removal on bound volumes
	Fine-tuned VLM	Modi, Kaithi and Urdu-nastaliq revenue hands — these are scripts, not scheduled languages, so no general OCR covers them
	Surya / Tesseract-Indic	Air-gapped fallback for states that forbid data egress
Geometry BUILT	Python 3.11	Same runtime as the existing backend
	shapely 2.1	Geometry predicates — intersection, union, distance, buffer. The same operations PostGIS exposes, so the port later is mechanical
	pyproj 3.7	The EPSG database and datum transforms. Registry is built from it at import so it cannot drift
	Pure Python optimiser	Deterministic compass search. Avoids a scipy dependency for a 2-parameter fit
API BUILT	FastAPI	Already in the project; Pydantic gives the OCR contract free validation and OpenAPI docs
	Pydantic v2	The schema <i>is</i> the contract between OCR and geometry
	Uvicorn	ASGI server
Frontend BUILT	Next.js 14 (App Router)	Existing project framework
	React 18 + TypeScript strict	Existing; types mirror the API response exactly
	Leaflet 1.9 + react-leaflet 4.2	Lightweight, no API key, OSM and Esri satellite basemaps. Loaded via <code>dynamic(ssr:false)</code> — Leaflet touches <code>window</code> at import
	Tailwind	Existing design system
Persistence NEXT	PostgreSQL + PostGIS	Supabase supports it. Needs <code>CREATE EXTENSION postgis</code> and a <code>geometry(Polygon,4326)</code> column
	Object store	Scanned pages with WORM / object lock for tamper evidence
	Async queue + workers	A 300-page bound volume cannot OCR inside a request

8.1 Migration to PostGIS

The engine was written as a thin wrapper over the same predicates PostGIS exposes, so the port is mechanical rather than a rewrite:

```

-- area reconciliation
SELECT survey_no,
       ST_Area(geom::geography)/10000
       AS mapped_ha
FROM   parcels
WHERE  ABS(ST_Area(geom::geography)/10000
          - recorded_area_ha)
       / recorded_area_ha > 0.05;

-- overlapping ownership
SELECT a.survey_no, b.survey_no,
       ST_Area(ST_Intersection(a.geom,b.geom)
              ::geography)/10000 AS overlap_ha
FROM   parcels a JOIN parcels b
       ON ST_Overlaps(a.geom, b.geom)
       AND a.id < b.id;

-- parcel plotting outside its own village
SELECT p.survey_no FROM parcels p JOIN villages v USING (lgd_village_code)
WHERE  NOT ST_Within(p.geom, v.geom);

```

SECTION 9

Ideal · current · implemented

CAPABILITY	HOW IT SHOULD IDEALLY WORK	HOW IT WORKS NOW	STATUS
Getting measurements off the page	Sarvam OCR reads the 7/12 and the tippan; a layout model parses the ladder diagram into chainage/offset rows; every field carries confidence	Documents are supplied as JSON — by hand, by fixture, or by upload	NOT BUILT
Bearings & distances	Any notation, magnetic or true, any unit	Whole-circle and quadrantal both parse and round-trip; 17 linear units; declination applied when supplied	DONE
Area units	Every regional unit resolved against the district's own revenue manual	19 standard units exact; bigha/katha keyed by state with confidence labels, and refused rather than guessed when the state is unknown	PARTIAL
Coordinate systems	Every CRS any Indian sheet was ever drawn on	14 India zones, 6 UTM zones, India NSF LCC; per-parcel zone selection; legacy zone inference; ambiguity refused	DONE
Shape from measurements	Traverse, chain-and-offset, and coordinates, all with self-checks	All three, with Bowditch/transit adjustment and closure grading	DONE
Position on Earth	Every parcel anchored to survey control	Coordinates, or Helmert fit to control points, or adjacency inference from one anchor	DONE
Village assembly	Least-squares network adjustment over the whole village simultaneously	Greedy growth from anchors plus iterative refinement; centimetre-exact on regular tiling, metre-scale on irregular	PARTIAL
Uncertainty	Rigorous covariance propagation	Empirical slack probe, propagated along the inference chain, conservatively calibrated	PARTIAL
Cadastral map sheets	Scanned village maps georeferenced with GCPs and raster-to-vector traced	Not addressed — the engine works from measurements, not from map images	NOT BUILT
ULPIN	Emit and consume the official 14-digit Bhu-Aadhaar	Not implemented in this layer. A demo derivation existed in the earlier build and was removed rather than left to be mistaken for the real thing	NOT BUILT
Persistence	PostGIS with topology validation and version history	Stateless — every request recomputes from the documents supplied	NOT BUILT
Officer workflow	Review queue, correction, digital signature, audit trail	Read-only map with provenance and findings	PARTIAL

What this means for the demo

The claim that can be made honestly is: *"Give us the measurements off any Indian land document and we will produce a validated, provenance-tracked polygon in WGS84, anywhere in the country, with an error bar. One anchored parcel maps an entire village."* The claim that cannot yet be made is that OCR produces those measurements automatically, or that scanned cadastral map sheets are vectorised.

SECTION 10

What to build next, in order

#	WORK	WHY IT IS NEXT	EFFORT
1	Tippan / FMB extraction to the JSON contract	The single blocker. Everything downstream is built and tested; without this the corpus cannot enter the system. Start with printed FMB tables before hand-drawn ladders	High
2	Officer anchoring UI — click two corners on satellite imagery	Converts a floating village into a mapped one in about thirty seconds, and needs no new backend. Highest value per hour of work in the whole list	Low
3	PostGIS persistence	Makes results durable and moves the validation rules into SQL where they scale across districts	Medium
4	Least-squares network adjustment	Replaces greedy growth with a simultaneous solve; would close most of the gap between the 0.025 m and 16 m results	Medium
5	Cadastral map sheet georeferencing	The other half of the problem statement, and the part DILRMP is genuinely stuck on. Deliberately separate from this engine	High
6	ULPIN emit and consume	Do it properly against the state's issuing service, or not at all	Low

Two things to keep saying out loud

Geometry is a second validator

It is not a map viewer. A traverse that fails to close catches a dropped digit that no OCR confidence score can see, because the error is only visible when the measurements are required to be self-consistent. The register's declared area and the reconstructed area are two independent statements about one parcel; when they disagree, something on the page is wrong.

Nothing here changes a legal record

Indian land title is **presumptive**, not conclusive. An inferred position is a hypothesis for an officer to verify, never a statement about who owns what. That is precisely why the accuracy class, the provenance and the uncertainty travel with every parcel through the API and onto the map — so the two can never be confused.

Engine: 2,983 lines across five modules, plus a 449-line API and a ground-truth test suite. Dependencies: shapely and pyproj. All figures in this report are measured outputs from the committed code, reproducible with `python tests/test_assembly_truth.py`. Repository: `D:\land\brain-land`.